

VELU AENEAS JERRON

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PROFESSIONAL SUMMARY

Hands on maker, researcher, and educator with a strong background in Physics and Computer Science, academic writing, and entrepreneurial skills. Experienced in innovation projects, digital fabrication, and applied research. Proven track record in training, mentoring, and managing technical projects in cross-disciplinary and international settings.

EDUCATION

MPhil in Physics – University of Jaffna, Sri Lanka (Aug 2021 – Jul 2025)

Thesis: *Simulation Study on Degradation of Scaffolds in Bone Tissue Engineering*

Research Exchange: Western Norway University of Applied Sciences, Bergen, Norway (Feb 2022 – Aug 2023) – Worked on numerical simulations (Finite Difference & Finite Element Methods) for biopolymeric medical implants.

BSc in Physics – University of Jaffna, Sri Lanka (Apr 2014 – May 2018)

Primary Subjects: *Physics, Computer Science, and Applied Mathematics*

Undergraduate Research Project: *Identifying Lung Cancer Using CT Image by Image Processing*

Industrial Visits: SLINTEC, Homagama (2015); Noritake Lanka Porcelain (PVT) Ltd, Matale (2015)

Clinical Visit: Tellipalai Trail Cancer Hospital, Jaffna (2017)

EXPERIENCE

Information Management Associate for Shelter, Land, and Site Coordination (SLSC) Sector – International Organization for Migration (UN Migration) (Feb 2026 – Jun 2026)

- Update the ShelterCluster website for Sri Lanka Cyclone Ditwah
- Coordination with Sector partners (Government, NGOs, and CSOs).
- Data management, Analysis, and Reporting.
- Kobo Tools development for the assessment.
- Supported DTM with tool development and Data cleaning.

Data Analyst – Institute for Development Research & Training (Jul 2024 – Jun 2025)

- Managed and analyzed project-related datasets for research and development programs.

Research Assistant – University of Jaffna (2021 – 2024)

- Conducted numerical and simulation studies for biodegradable medical implants.
- Prepared research documentation and coordinated academic writing tasks.

Teaching Assistant – University of Jaffna & Open University of Sri Lanka (Jun 2018 – Jan 2021)

- Conducted Physics laboratory demonstrations, evaluated reports, and supported course delivery.

Teacher/tutor: Mathematics , Physics, Chemistry and Biology for IGCSE students (2013 – To date)

Workshop & Training Facilitator (2019 –2021)

- Delivered technical workshops in Arduino, PCB design, MATLAB/SIMULINK, IoT, and innovation for undergraduate students and educators.
- Conducted Makerspace program for school students at American Center Colombo.

PUBLICATIONS & CONFERENCES

- **Velu Aeneas Jerron**, Thirunavukkarasu Pathmathas, Talal Rahman, Dhayalan Velauthapillai; A Unified Multiphysics model for Degradation of bone Tissue Engineering Scaffolds; International Conference on Advanced Materials for Clean Energy and Health Applications, Coimbatore Institute of Technology, India, June 03-05, 2026.
- **Aeneas Jerron Velu**, Pathmathas Thirunavukkarasu, Talal Rahman, Kamal Mustafa, Dhayalan Velauthapillai; Computational Analysis of Biodegradable Polyester Materials for Biomedical Applications: Investigating Molecular Weight Change due to Hydrolysis, ACS Omega 2024 9 (17), 19108-19116 DOI: 10.1021/acsomega.3c10112.
- **Aeneas Jerron Velu**, Pathmathas Thirunavukkarasu, Dhayalan Velauthapillai; Modeling Bone Tissue Growth and Scaffold Degradation Using Wolff's Law and Reaction-Diffusion Dynamics; Advanced Materials for Clean Energy and Health Applications, University of Jaffna, Sri Lanka, March 27-28, 2025.
- Saseekaran Apirakshan, Anton Gratian Desman, **Aeneas Jerron Velu**, Pathmathas Thirunavukkarasu, Dhayalan Velauthapillai; Exploring the Effect of Temperature on Poly (Lactic-co-Caprolactone) Degradation: A Comparative Analysis; Advanced Materials for Clean Energy and Health Applications, March 27-28, 2025.
- Sripathmanathan Anurakavan, **Aeneas Jerron Velu**; Portable Solar-Powered Water Purification and Hydrogen Production System: Integrating Solar Still and Water Electrolysis in a Compact Unit; Advanced Materials for Clean Energy and Health Applications, March 27-28, 2025.
- **Aeneas Jerron Velu**, Pathmathas Thirunavukkarasu, Dhayalan Velauthapillai; Mechanical evolution of tissue engineering scaffolds: Simulating semi-crystalline polymer degradation, Vingnanam Research Conference, September 06, 2024.
- **V A Jerron**, S Anurakavan; Smart Tracking of Plastic Containers for An Efficient Waste Management Future, International Conference on Plastics & Environmental Sustainability, March 07-08, 2024.
- S Anurakavan, B C L Athapattu, **V A Jerron**; Gamification of Plastic Waste Management for Behavioral Change, International Conference on Plastics & Environmental Sustainability, March 07-08, 2024.
- **Aeneas Jerron Velu**, Pathmathas Thirunavukkarasu, Talal Rahman, Dhayalan Velauthapillai; In Silico Study on Biodegradation and Crystallization of Polymeric Implants; International Conference on Advanced Materials for Clean Energy and Health Applications, Coimbatore Institute of Technology, India, September 19-20, 2023.
- **Aeneas Jerron Velu**, Pathmathas Thirunavukkarasu, Dhayalan Velauthapillai; Computational Modeling and Simulation Study on Degradation of Scaffolds in Bone Tissue Engineering; Technological Advances in Science, Medicine and Engineering Conference, July 03-04, 2021.

SELECTED PROJECTS & ACHIEVEMENTS

- **Solar-powered water purification system** – Presented at International Conference (2025)
- **Smart plastic tracking system** – Presented at International Conference (2024)
- Youth Forum Member, American Corner Kandy (2021–2022) – Led English-speaking forum
- Participant, **National Youth Social Innovation Challenge (2018)** – Learned design thinking
- **Reverse vending machine for PET bottle collection** – 1st Runner-up, Sampath Green Inventor (2018)

CERTIFICATIONS & TRAINING

- **LK Domain Registry** – Database Management Systems & SQL (2026)
- **WIPO Academy** – IP Training Strategies (2025)
- **Agriculture.lk** – Professional Certificate in Human Resource Management (2025)
- **Google (Coursera)** – Introduction to AI (2025), Maximize Productivity With AI Tools (2025), Discover the Art of Prompting (2025), Use AI Responsibly (2025), Stay Ahead of the AI Curve (2025), Foundations of Project Management (2025)
- **KoboToolbox** – Humanitarian Needs Assessment (2025)
- **Harvard (edX)** – Introduction to Probability (2024)
- **UoM CODL** – Python for Beginners (2023)
- **deeplearning.ai (Coursera)** – Neural Networks, Project Structuring, Optimization (2022)
- **University of Michigan (Coursera)** – Python Data Structures (2022)
- **Duke (Coursera)** – Data Science Math Skills (2022)
- **Harvard (edX)** – Using Python for Research (2021)
- **American Corner Kandy** – Cybersecurity, Business English, DynEd (2021)
- **American Center Colombo** – Makerspace Training (2018)
- **Leadership Program** – Ministry of Higher Education (2013)

TECHNICAL SKILLS

- **AI:** Prompt Engineering, Gen AI Tools, AI assisted productivity
- **Digital Fabrication:** PCB design, Arduino projects, microcontroller programming
- **Software & Tools:** MATLAB, Python, C++, FORTRAN, Java, PHP
- **Research & Analysis:** Simulation methods, data analysis, academic writing
- **IT Proficiency:** Windows, Linux/Unix, web development basics

REFERENCES

Dr. T. Pathmathas

Senior Lecturer
Former Head of Department
Department of Physics
University of Jaffna

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